

- **Prompt Engineering**
 - Basic and advanced prompting, prompt evaluation
- **LLM-Based Agents**
 - Agent concepts, architectures, reasoning and planning
 - Tool calling, orchestration and multi-agent collaboration
- **Knowledge, Context and Memory**
 - RAG, context management and agent memory
- **Evaluation and Assurance**
 - Evaluation and benchmarking
 - Reliability, bias, privacy, safety and ethical considerations

Recommend readings

- See list on loop
- Additional references will be provided in lecture slides
- **O'Reilly** – several books on LLMs and LLM-based agents available through DCU access.
- **Key references**
 - Alammam, J. & Grootendorst, M. (2024). *Hands-On Large Language Models: Language Understanding and Generation*. O'Reilly Media. 15 October 2024.
 - Phoenix, J. & Taylor, M. (2024). *Prompt Engineering for Generative AI: Future-Proof Inputs for Reliable AI Outputs*. O'Reilly Media. 31 May 2024.
 - Hunter, N. (2023). *The Art of Prompt Engineering with ChatGPT: A Hands-On Guide*. Independently published. 19 January 2023.
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Staff and structure

- **Lecturer** - Prof. Boualem Benatallah (Full Professor, School of Computing, DCU, Fellow of the IEEE)

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<https://dblp.org/pid/b/BoualemBenatallah.html>,

<https://scholar.google.com/citations?user=WYmChp0AAAAJ&hl=en>

- **Structure**

- Lectures : Room (GLA.QG15), Monday : 17:00 – 18:00. Room may change, so check announcements regularly.
- Lecture/tutorials: Room (GLA.HG20), Wednesday : 09:00 – 10:00. Room may change, so check announcements regularly.
- Materials (slides, activity materials and guidelines) will be available in Loop, Use of Loop, DCU email for module communication

- **50% Continuous Assessment**
 - Group project
- **50% final exam**
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Learning outcomes

- Explain the core concepts and architectures of LLM-based agents, including prompt engineering, retrieval-augmented generation, memory and context management, task decomposition, tool calling, safeguards, and evaluation.
- Design and analyse agent solutions, including agent orchestration and multi-agent collaboration, for a given task.
- Select and evaluate agent frameworks for applied or research-oriented use cases.
- Evaluate agent behaviour and outputs using appropriate methodologies, including benchmarks.
- Apply quality assurance approaches for agent systems, including robustness and reliability checks, safety, and ethical considerations.
- Integrate tools, memory, orchestration strategies, and evaluation components into an end-to-end AI agent system addressing a realistic task

Communication

- Announcements: DCU email, Loop
- Slides will be added to each topic as we cover different parts
- Guidelines for group activities and marking schemes for the various project parts will be provided in due time.
- Students are required to form groups, preferably 4/5 students per group. Instructions will be sent after first lecture
- There will be consultation hours every week during the semester starting Week 2 (schedule will be made available in Loop)
- You can also schedule a consultation (please send an e-mail to my DCU email, boualem.benatallah@dcu.ie)